Agent-Computer Observation Interfaces Enable Dynamic Computer Use

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Abstract

SWE-agent [Yang et al., 2024] established the *action* interface as an underexplored design axis for software-engineering agents. We make the analogous case for the *observation* interface in computer-use (CU) agents. Current CU agents, both closed and open-source, tie observation to action: one screenshot every 3–5 s, with no audio. Between screenshots they are blind and deaf to video, animations, transient UI events, meetings, and spoken instructions.

We introduce the **Agent-Computer Observation Interface (AOI)**, a model-agnostic perception layer that decouples continuous, adaptive observation from discrete actions. It has three gated components: inter-step keyframe capture, volume-gated audio transcription, and CU-model-generated visual narration that persists as text. Each one produces almost nothing on static, silent content, reducing to the standard loop without degrading it.

On **DynaCU-Bench** (100 dynamic browser tasks plus a 50-task static control), eight CU models from 7B to frontier scale gain $+17$ to $+48$ pp over their screenshot baselines with zero retraining (paired McNemar $p < 10^{-3}$ for all). Claude Sonnet 4.6 reaches a 3-seed mean of 78.7%. On a 12-task audio subset, AOI agents score 12/12 where current-generation streaming voice models fail on their own. OpenAI's GA Realtime (`gpt-realtime-2`) reaches 2/12 and xAI's Grok Voice 1/12 despite accurate hearing, because they cannot ground actions without the AOI's scaffold (with which the Realtime model recovers to 11/12). Three findings reframe CU perception. First, keyframe *selection* is unimportant: value emerges only once captured frames are narrated into persistent text. Second, narration helps mainly by making the model *articulate* what it sees. Third, the interface is not a fixed bundle. On Gemini 3 Flash the keyframe stream regresses -12 pp from image-token dilution, so components must be tuned per model rather than shipped as one configuration.

Code: https://github.com/19PINE-AI/AOI
Website: https://01.me/research/AOI

1 Introduction

Computer-use agents are blind between screenshots and deaf to audio. A computer-use (CU) agent perceives the screen as a periodic stream of static screenshots captured every 3–5 seconds and has no audio channel at all (Figure 1). Between snapshots it is *blind*: videos play, slides animate, toast notifications appear and auto-dismiss, and all of this remains unobserved if the change does not coincide with a screenshot. Furthermore, it is *deaf*: meeting speeches,

spoken instructions, and notification chimes will never reach the model. An entire class of everyday computer use is therefore out of reach, even for agents that handle static interfaces well. This includes watching screencasts, following meetings, answering voice prompts, and reacting to transient dialogs.

The gap is real and unaddressed. On static GUI benchmarks such as OSWorld [Xie et al., 2024], screenshot agents have improved from sub-10% to above 50% sucess [Andreux et al., 2025, Wang et al., 2025a], approaching the human baseline in filling forms, navigating sites, editing documents, and operating professional software. However, the dynamic-and-audible regime is documented but largely unaddressed gap: video-LLM GUI agents struggle across temporal tasks [Chen et al., 2024], long-context models do *worse* with video input than without [Jang et al., 2024], paused-frame game evaluation reach only single-digit completion [Zhang et al., 2025], and screenshot agents miss dozens of hours of temporal dynamics [Chen et al., 2025]. Even a recent CU survey catalogues many open challenges but not the observation modality itself [Sager et al., 2025].

The observation interface is a separable design axis. Why is this hard to fix from the model side? Retraining CU models on video is expensive, model-specific, and undemonstrated at scale. Video-native models brute-force frame sampling and waste tokens on redundant static frames. Meanwhile, native real-time multimodal APIs [Google, 2025, OpenAI, 2024] work but lock users to one provider, offer no selective perception, and require migration (we compare against them in Section 8). We take a different approach, inspired by SWE-agent [Yang et al., 2024]. Rather than training a new model, SWE-agent demonstrated the power of redesigning the agent-computer *action* interface: what commands the model receives and how inputs and feedback are structured. We argue the *observation* interface is the analogous lever for CU agents. Prior work focused on enriching *what* a single snapshot encodes, such as accessibility trees, element overlays, and text surrogates. It left untouched the dimension we target: *when* the agent looks and through which senses. Our core principle is *decoupling observation (continuous, adaptive, multimodal) from action (discrete)*. Every existing CU agent we are aware of [Anthropic, 2024a, OpenAI, 2025, Google DeepMind, 2025, Wang et al., 2025a,b, Awadallah et al., 2025, Song et al., 2025] ties the two together at one static screenshot per step while remaining completely deaf to audio.

Our approach. The **Agent-Computer Observation Interface (AOI)** is a lightweight perception layer placed between the environment and any image-based CU model. It converts continuous screen and audio streams into the sparse images and text the model already accepts through three gated components: inter-step keyframe capture, volume-gated audio transcription, and model-generated visual narration that persists as text after the source images are pruned. On static, silent content the system produces almost no additional input and the behaviour reverts to the standard loop. On dynamic or spoken content, the model additionally sees what moved and hears what was said. No retraining is required, although the optimal component mix is model-specific.

Contributions.

1. We identify the **observation interface** as a separable CU design axis, complementary to SWE-agent's action interface, with *decoupling observation from action* as the core principle (Sections 1–3).
2. We introduce the **Agent-Computer Observation Interface (AOI)**, a model-agnostic perception layer with three gated components. We releast AOI together with **DynaCU-Bench**, a

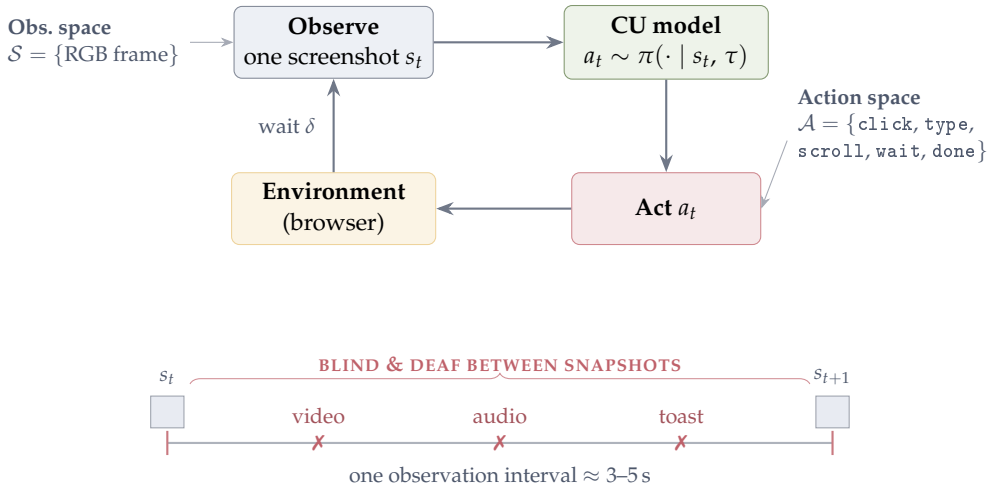


Figure 1. The computer-use agent loop and its blind spots. A CU agent repeats *observe* → *reason* → *act*: it captures a single screenshot s_t (observation space $\mathcal{S}$), the model samples an action a_t from a small grammar $\mathcal{A}$ (`click/type/scroll/wait/done`), the browser executes it, and after a buffer δ the next screenshot is taken. The interval is 3–5 s; *between* snapshots the agent is blind to anything that moves (video, animation, an auto-dismissing toast) and has no audio channel at all, so spoken content is never observed.

benchmark of 100 dynamic tasks across 10 categories, and a static task set for degradation control.

3. Across eight CU models from 7B to frontier scale, the AOI delivers significant gains on dynamic tasks with zero retraining (Section 5), while staying transparent on static work.
4. We provide a detailed decomposition of the gains: keyframe *selection* is largely irrelevant, most value emerges only once frames are narrated into persistent text, and the component mix must be tuned per model, with one component even becoming harmful on a newer model (Sections 6.2–7).

2 Background: The CU Agent Loop

The loop. Every current CU agent runs the same *observe* → *reason* → *act* cycle shown in Figure 1: it captures a screenshot s_t, samples an action a_t from the model conditioned on s_t and trajectory history τ, executes the action, waits a short buffer δ, and repeats. Two spaces define an agent's capabilities. The **action space** $\mathcal{A}$ is a small grammar of discrete commands (click, type, scroll, wait, done) and has been the focus of prior interface work [Yang et al., 2024]. In contrast, the **observation space** $\mathcal{S}$, the focus of this work, remains limited: today it consists of a single RGB frame, $\mathcal{S} = \{\text{one screenshot}\}$, sampled once per action.

Why one screenshot per step leaves the agent blind. The observation interval (model inference, action execution, and buffer) typically lasts 3–5 seconds, and each screenshot is a single point sample per interval. Consequently, anything that happens *between* samples is lost: the captured frame is static, rendering motion invisible, and there is no audio channel so sound is never captured. Multi-image histories do not help. The extra images are the post-action screenshots from prior steps and never the moments between them. Concretely, four categories of content are systematically missed. (i) *Transient events* are toasts and dialogs that appear and auto-dismiss within a single interval. (ii) *Continuous visual media* are video, animated tutorials,

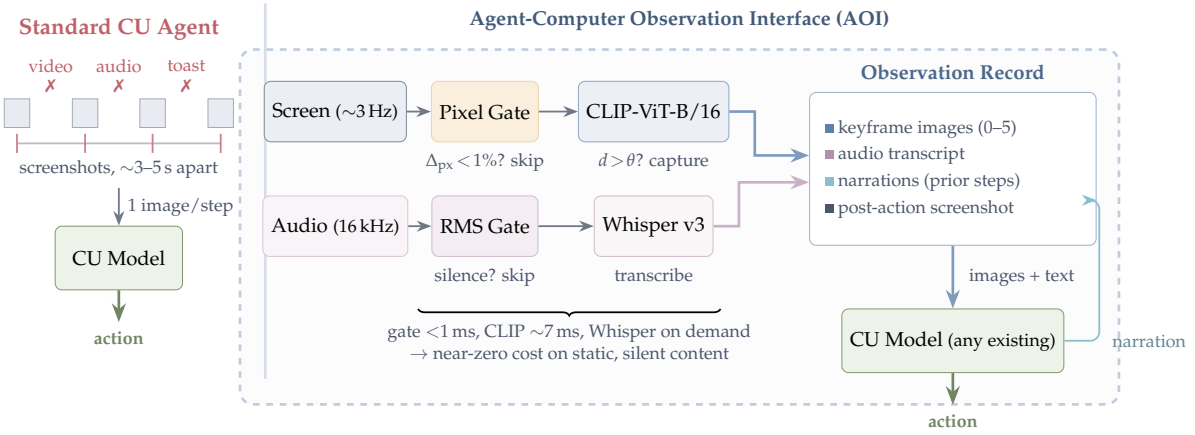


Figure 2. Standard CU agents vs. AOI-equipped agents. *Left:* Current agents observe through periodic screenshots (∼3–5 s apart) and are blind and deaf between observations, missing video, audio, and transient UI events (✗). *Right:* The AOI continuously monitors screen and audio streams through fast gates (<1 ms) that skip processing for static, silent content. When gates fire, keyframe extraction (shown here with optional CLIP filtering, see Section 6.2) and Whisper ASR produce images and text for the observation record. Visual narrations feed back as persistent text memory. The AOI wraps any existing CU model with zero retraining.

and transitioning slides. (iii) *Audio* is meeting speech, notification sounds, and spoken prompts. (iv) *Periodic noise* is spinners and blinking cursors that should be *ignored* yet disrupt naive frame differencing methods. AOI expands $\mathcal{S}$ to cover (i)–(iii) while suppressing (iv), all without touching the model or $\mathcal{A}$.

3 System Design

The AOI is inserted between the environment and any existing CU model. It continuously observes the entire interval between agent steps and provides the model with additional keyframes, audio descriptions, and accumulated visual narration, all in the standard image + text format that every CU model already accepts. On static, silent tasks, the AOI's perception gates remain mostly closed and behaviour reduces to the standard loop wrapped in AOI's structured observation-record format (which does not degrade static work, and Section 6.1 shows it slightly helps). On dynamic tasks, the model additionally receives the inter-step keyframes, audio transcripts, and narration. This design provides strictly more *information*. However, it does not always provide a strictly better *outcome*: the per-step observation work adds latency, which can outweigh potential gains on action-bound tasks (browser games, Section 5). The AOI targets human-paced computer use, not real-time interaction. No model retraining is needed.

3.1 Architecture Overview

The AOI consists of three components, each preceded by a fast (sub-millisecond) gate that short-circuits processing when there is nothing new to perceive. Figure 2 contrasts the resulting pipeline with the standard CU loop. For frames that pass the gates, the additional cost is dominated by CLIP-ViT-B/16 (∼7 ms on GPU) for visual frames and Whisper large-v3 (variable, on demand) for audio segments. For the static, silent frames that dominate most workloads,

Algorithm 1 Two-stage adaptive keyframe extraction (CLIP variant). This algorithm is shown for completeness. The selection-method ablation in Section 6.2 finds it statistically indistinguishable from the simpler pixel-only gate (Stage 1) and uniform / random sampling, so the default selector in every main result is Stage 1 alone.

Require: Screen sample x_t, anchor embedding e_{anchor}, pixel threshold α, CLIP threshold θ
1: $\Delta_{\text{px}} \leftarrow \text{PixelChangeRatio}(x_t, x_{\text{prev}})$
2: **if** $\Delta_{\text{px}} < \alpha$ **then**
3: **return** SKIP {Stage 1: no pixel change}
4: **end if**
5: $e_t \leftarrow \text{CLIP}(x_t)$
6: $d \leftarrow 1 - \cos(e_t, e_{\text{anchor}})$
7: **if** $d > \theta$ **then**
8: $e_{\text{anchor}} \leftarrow e_t$ {Re-anchor}
9: **return** CAPTUREKEYFRAME(x_t, t)
10: **end if**
11: **return** SKIP {Stage 2: below semantic threshold}

the per-frame cost reduces to the sub-millisecond gate alone.

(1) **Inter-step keyframe capture**: Continuous screen sampling at ∼3 Hz with gating to suppress redundant frames. Produces 0–5 keyframe images per step.
(2) **Volume-gated audio observer**: RMS energy gate followed by ASR (Whisper large-v3). Produces a text transcription per step, only when audio is present.
(3) **Visual narration context**: The CU model itself generates a brief text description of new visual information as a side-output of each inference call. These narrations accumulate in the trajectory and persist after keyframe images are pruned from context.

3.2 Inter-Step Keyframe Capture

Between agent steps, the screen may change (a video plays, a dialog appears, a slide transitions) or stay static. The AOI continuously samples the screen at ∼3 Hz and selects a subset of frames to include as keyframe images in the observation record (up to 5 per step). Multiple selection strategies are possible: uniform fixed-rate sampling, pixel-level differencing, random sampling, or semantic filtering via CLIP embeddings [Radford et al., 2021]. We evaluate five variants spanning these four families (uniform at 1 and 3 FPS, pixel-diff, random, CLIP) in Section 6.2 and find that all converge to similar accuracy, so the default implementation uses simple pixel-change gating: if fewer than 1% of pixels changed since the last captured frame, the sample is skipped (cost: <1 ms on CPU). For static screens, no keyframes are produced and the component has zero overhead. Algorithm 1 formalises the two-stage CLIP variant. The default selector drops the CLIP stage (lines 5–10) and uses only Stage 1.

3.3 Volume-Gated Audio Observation

Current CU agents have zero audio perception. They cannot hear meeting speech, notification sounds, or error alerts. Pure ASR models (Whisper) transcribe speech but produce nothing for non-speech audio events.

Volume gate. In most computer use, such as file editing, form filling, and silent web

browsing, there is no audio. Computing RMS energy of the audio buffer takes sub-millisecond time. If the energy falls below a silence threshold, the full ASR call is skipped entirely. This yields ∼0 ms additional cost for the majority of steps.

When audio is present, we transcribe speech with Whisper large-v3 [Radford et al., 2023], operating on 16 kHz mono audio captured from the system audio output via PulseAudio virtual devices. Our implementation focuses on transcribing *speech* only. Non-speech audio events (notification chimes, error beeps) may trip the volume gate but will produce no useful transcript. Consequently, the audio channel's contribution throughout this paper is spoken content only. A multimodal audio model would be needed to interpret the full audio scene (Section 10). The audio buffer spans the full inter-step interval, including the post-action buffer. This ensures speech that begins just after an action (e.g. a spoken confirmation) is still captured.

Overlapping windows. Agent step boundaries are determined by model inference latency rather by speech content. As a result, a typical 3.5-second audio chunk, which captures ∼8–9 words at normal speaking rate, will almost always fall mid-sentence. To resolve these boundary artifacts, the audio buffer includes ∼3.5 seconds of overlap from the previous interval to provide acoustic continuity across step boundaries.

3.4 Visual Narration for Long-Term Context

CU models retain only the most recent images in context, pruning earlier keyframes. Consequently, on tasks that span many steps, the agent loses all visual memory of prior content. While audio transcriptions persist naturally as text in the trajectory, visual observations have no analogous persistence.

To address this, at each step the CU model outputs both an action and a brief visual narration, a text description of what is visually new in the current keyframes. This narration is generated in the same inference call that produces the action, adding no extra inference call and only a small number of output tokens. It persists indefinitely in the trajectory, even after the corresponding images are pruned. Following the caption-then-reason principle of LLoVi [Zhang et al., 2023], narrations are generated by the CU model itself (no separate captioner) and are task-relevant rather than generic.

3.5 Observation Record

At each step, the CU model receives a structured document combining text context from recent prior steps (audio transcriptions, visual narrations, actions taken) and raw observations from the current interval (new audio transcription, any captured keyframe images, and the post-action screenshot). For static, silent tasks, the raw-observation part of this document reduces to just the post-action screenshot. However, the surrounding structured format (DOM element list, prior-step text trajectory) is still present. As a result, the input is not identical to a raw-screenshot loop. Furthermore, Section 6.1 shows this format alone gives a small gain on purely static work. Figure 3 shows a concrete example of the document format and Figure 4 grounds the pipeline in real pixels and verbatim narrations from a recorded benchmark run.

3.6 Implementation

The AOI is a modular Python layer (∼2,600 lines of code) that wraps any CU model. Screen and audio are captured through a headless browser and virtual audio devices. The keyframe

Observation Record — Step N (Meeting task)

```
# CONTEXT -- text from prior steps (no images)
Step N-1 (t ≈ 18.5-22.0 s):
AUDIO: "Here's the team.  As you can see we
have strong cross-functional coverage."
VISUAL: "Meeting slide shows Project Team
with 6 members listed in a grid."
ACTION: wait()
# NEW -- current interval observations
Step N (t ≈ 22.0-25.5 s):
AUDIO: "And I want to mention, we've moved
the launch date to April 28th.
Please update your calendars."
[IMAGE: post-action screenshot]
# TASK
Read the meeting slides and listen to the
discussion.  What is the new launch date?
Enter it in the Launch Date field.
```

Figure 3. Example observation record sent to the CU model at step N of a meeting task. The **context** section provides text from prior steps (audio transcriptions and visual narrations persist after images are pruned). The **new** section contains the current audio transcription and the post-action screenshot (image). In this case, no keyframes were captured (the slide did not change), so only the screenshot is included as an image. The task instruction is appended at the end.

encoder runs on the GPU and speech transcription is handled by a separate CPU service to avoid contention with the model. Cloud models are called through their native APIs and local models are served on a single GPU. Full software, hardware, and hyperparameter details are in Appendix I.

4 DynaCU-Bench

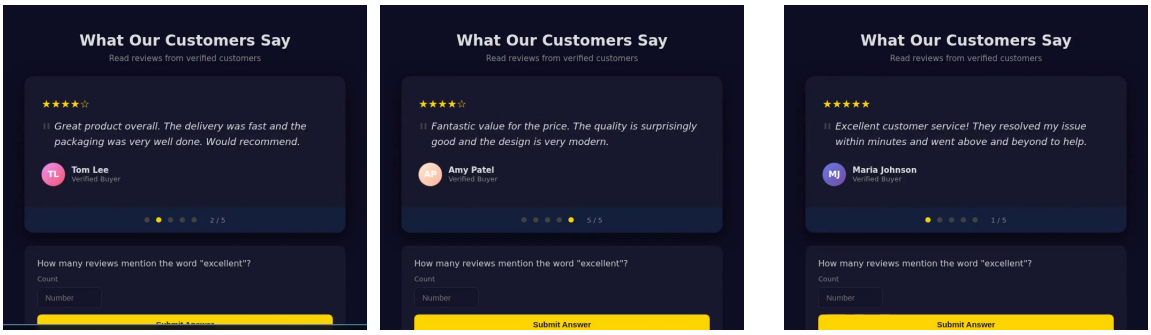
4.1 Design Principles

Existing CU benchmarks [Xie et al., 2024, Zhou et al., 2023, He et al., 2024] consist almost entirely of tasks solvable from static screenshots. We introduce **DynaCU-Bench**, a benchmark of 100 browser-based tasks that specifically require dynamic visual and/or audio perception. These tasks are unsolvable from static screenshots alone.

Design principles: (i) Tasks must *require* temporal visual or audio perception. (ii) Tasks must represent realistic computer-use scenarios, not artificial constructs. (iii) Tasks can be executed in a reproducible headless browser environment. (iv) The benchmark covers all four dynamic content categories from Section 2. (v) The benchmark includes difficulty stratification (3 easy, 4 medium, 3 hard per category).

4.2 Task Categories

DynaCU-Bench spans 10 task families across three capability axes: audio perception (AUD), visual-temporal perception (VIS), and real-time interaction (INT). Throughout the paper, we name each family descriptively (Table 1), and use these names consistently in every table and figure. Each task family has 10 tasks (3 easy, 4 medium, 3 hard). Representative examples



step 1: testimonial 2/5 (Tom Lee) step 2: testimonial 5/5 (Amy Patel) step 3: testimonial 1/5 (Maria Johnson)

Visual narrations accumulated in the trajectory (verbatim from the evaluation log)

Step 1 (0 keyframes): "The carousel shows testimonial 2/5 from Tom Lee with a 4-star rating about fast delivery and good packaging, with no mention of 'excellent'." ACTION: click(567, 400)

Step 2 (1 keyframe): "The carousel now shows testimonial 5/5 from Amy Patel with a 4-star rating about fantastic value, quality, and modern design -- no mention of 'excellent'." ACTION: click(567, 401)

Step 3 (2 keyframes): "The carousel has cycled back to testimonial 1/5 from Maria Johnson with 'Excellent customer service!' -- and the transition keyframe also shows a Sarah Robinson testimonial partially visible..." ACTION: fill("2") ✓

Figure 4. What the agent actually sees: real frames and narrations from a recorded benchmark run (Carousel task: "How many reviews mention the word 'excellent'?" Claude Sonnet 4.6 + AOI full, the run reported in Table 2, passed in 3 steps.) The testimonial carousel rotates every few seconds, so no single screenshot can show more than one of the five reviews. *Top:* screen states observed across the three agent steps (frames from the run recording, recorder status bar cropped). *Bottom:* the visual narrations the CU model generated at each step, quoted verbatim from the evaluation log. The narrations persist as text in the trajectory, letting the agent accumulate evidence across carousel states and answer "2" at step 3.

appear in Appendix F.

4.3 Evaluation Protocol

Each task runs in a headless Chromium browser with virtual audio devices for playback and microphone injection. Spoken content is rendered using high-quality text-to-speech. Most tasks (93) are scored by a deterministic check on the resulting page state. The remaining tasks either combine that check with an LLM quality rubric (6) or rely solely on the LLM judge (1). A run ends after 15 agent steps or a per-task wall-clock budget (the stimulus duration plus a fixed margin and any AOI overhead), whichever occurs first. These budgets bind only for the slowest models (e.g. Grok-4 at ∼80–180 s/call, Section 7).

5 Evaluation

5.1 Setup

CU models. We evaluate six models spanning the capability and scale spectrum. (i) Claude Sonnet 4.6 [Anthropic, 2024a] is closed-source. (ii) GPT-5.4 [OpenAI, 2026] is closed-source, the

Table 1. DynaCU-Bench task families. Axes: AUD = audio perception, VIS = visual-temporal perception, INT = real-time interaction. Each family contains 3 easy, 4 medium, and 3 hard tasks.

Family	Axes	Description
Podcast	AUD	Listen to spoken narration, then extract facts or answer questions
Meeting	AUD+VIS+INT	Follow a meeting with slides and speech, then take notes or act
Screencast	VIS	Watch terminal/IDE recordings with auto-advancing frames
Carousel	VIS	Perceive CSS transitions, rotating content, product carousels
Dashboard	VIS	Monitor live-updating dashboards
Transient	VIS	Respond to toasts and auto-dismissing banners
Phone	AUD+INT	Listen to a synthesized voice call and respond
Interview	AUD+INT	Answer spoken questions and complete interviews
Collab	VIS+INT	Handle remote edits in collaborative documents
Games	VIS+INT	Play interactive games requiring temporal attention

latest GPT-5 series tool-calling-capable vision model exposed by OpenAI's chat-completions API at evaluation time. (iii) Gemini 2.5 Flash [Google DeepMind, 2025] is closed-source. (iv) Grok-4 [xAI, 2026] is closed-source (xAI), reported in Section 7 with attention to the inference-latency confound. (v) EvoCUA-32B [Meituan, 2026] is open-source (32B), with weights and inference recipe released through Meituan's GitHub. (vi) Fara-7B [Awadallah et al., 2025] is open-source (7B). In addition, we evaluate two further open-source models accessed through OpenRouter: Qwen3-VL-235B-A22B-Instruct and Qwen3-VL-30B-A3B-Instruct [Qwen Team, 2025]. These serve as a replication on the standard vs. AOI full contrast (Table 5). We use each model's vendor-recommended decoding settings (temperature, top-p) without modification. All reported results are measured directly in our DynaCU-Bench harness. We do not rely on any vendor-reported scores.

Observation configurations. The primary contrast is between **Standard** (one screenshot per step and no audio, the current paradigm) and **AOI full** (inter-step keyframes + audio transcription + visual narration). We evaluate both modes across all six models. All remaining modes are ablations and diagnostic controls run on Claude Sonnet 4.6 (unless otherwise noted). A glossary of every mode used in the paper is in Appendix H.

5.2 Main Results

Table 2 and Figure 5 present the main results: AOI delivers large, consistent gains across all six models, with every improvement statistically significant by paired McNemar's exact test ($p < 10^{-3}$). *Statistical protocol.* Each configuration is a single trial of 100 binary tasks (no repeats). We report 95% Wilson confidence intervals and evaluate differences with paired McNemar exact mid-p tests on the discordant tasks. This design applies to every per-family and ablation cell in the paper. Differences within the $\sim$3 pp decoding-noise band quantified

Table 2. DynaCU-Bench results: task success rate (%) on the 100 dynamic tasks. Brackets show 95% Wilson confidence intervals. **Bold** indicates the best result per model. Δ is the absolute gain over the standard baseline. *p*-values are paired McNemar's exact mid-*p* tests. "Claude 4.6" abbreviates Claude Sonnet 4.6 throughout. Grok-4 is the slowest model at inference (∼80–180 s/call), which holds its absolute scores down even when AOI unblocks perception (see Section 7).

Configuration	Claude 4.6	GPT-5.4	Gemini 2.5	Grok-4	EvoCUA-32B	Fara-7B
Standard (screenshot)	38 [29.1, 47.8]	37 [28.2, 46.8]	21 [14.2, 30.0]	4 [1.6, 9.8]	18 [11.7, 26.7]	17 [10.9, 25.5]
AOI (full)	**82** [73.3, 88.3]	**57** [47.2, 66.3]	**69** [59.4, 77.2]	**25** [17.5, 34.4]	**55** [45.2, 64.4]	**34** [25.5, 43.7]
Δ (full − std)	+44	+20	+48	+21	+37	+17
p (McNemar)	1.3×10^{-10}	8.8×10^{-5}	2.9×10^{-12}	3.0×10^{-6}	3.0×10^{-9}	4.9×10^{-4}

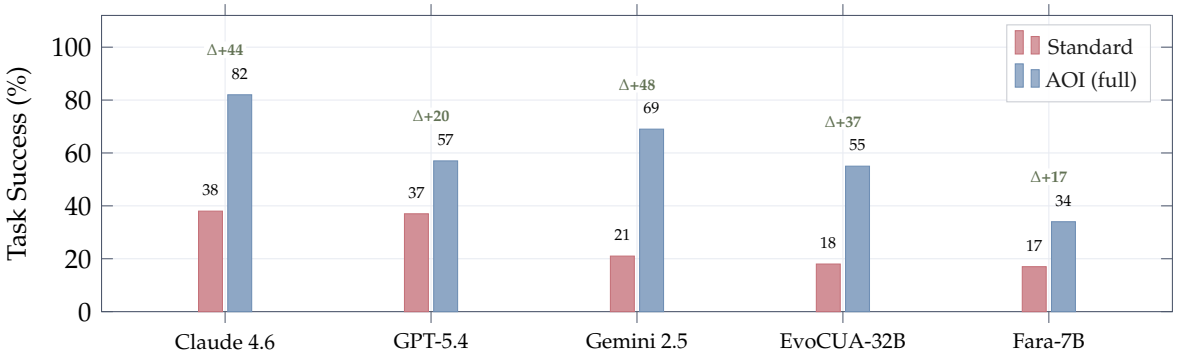


Figure 5. Main results on DynaCU-Bench (100 tasks). The AOI produces large gains across CU models without any retraining. Green numbers show absolute improvement in percentage points. Gemini 2.5 Flash shows the largest absolute gain (+48 pp) and relative gain (3.3×), while Claude achieves the highest absolute score (82% at seed 1, 3-seed mean 78.7%). Five of the six models tested are plotted here. Grok-4 (standard 4, AOI full 25, Δ+21 pp) is omitted from the chart for legibility and is reported separately in Section 7. Exact (asymmetric) 95% Wilson confidence intervals on every bar are tabulated in Table 2 and used for all reported *p*-values.

below should be read as noise unless a paired test says otherwise.

The per-model numbers (Table 2) carry three points beyond the raw gains. *Claude Sonnet 4.6* reaches the highest absolute score. Section 6 decomposes how its components combine. *GPT-5.4* gains as reliably as Claude but tops out lower, suggesting its CU capability, not perception, is the binding constraint once observation is adequate. *Gemini 2.5 Flash* posts the largest absolute gain despite being video-native: the standard agent loop feeds it only periodic screenshots, so its streaming architecture lies idle. Section 6 (Table 11) splits this into +25 pp from the structured prompt format and +23 pp from new inter-step perception. It benefits from *both*.

Among the open-source models, *EvoCUA-32B* (32B) barely functions on dynamic tasks without the AOI yet rises to rival GPT-5.4's AOI score, while *Fara-7B* (7B), the most reasoning-constrained, still benefits substantially. The pattern that gains track reasoning capacity recurs in the Qwen3-VL replication below.

Two side-by-side trajectories illustrate the mechanism (Figure 11, Appendix D). On a meeting task the standard agent exhausts its entire step budget because it cannot hear the spoken answer, while the AOI agent transcribes it and acts in two steps. On a screencast task the AOI agent's narration captures the answer in a single step.

Additional open-source replication (Qwen3-VL family). Two more open-source models

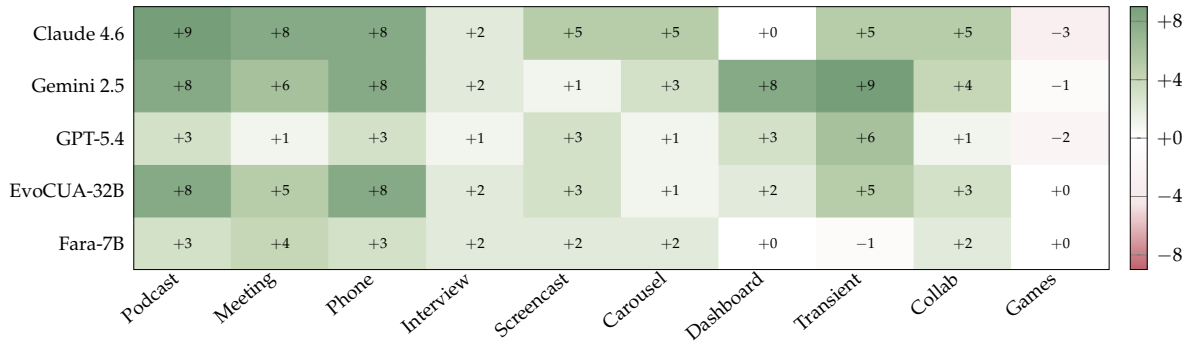


Figure 6. Per-family AOI gain (AOI full − Standard, in tasks per 10) across five models, families grouped by primary axis (audio, visual, interaction). Green = AOI helps, red = AOI hurts. Gains are positive almost everywhere and largest on the audio families (Podcast, Phone, Meeting), which are impossible from screenshots alone; the lone consistent regression is GAMES, where action latency—not perception—is the bottleneck. Absolute per-cell scores are in Appendix A.

run through the same contrast via OpenRouter both land inside the +17 to +48 pp band of Table 2 (full numbers in Appendix A, Table 5): Qwen3-VL-235B-A22B-Instruct [Qwen Team, 2025] improves +42 pp (22 → 64, surpassing GPT-5.4's AOI score of 57) and Qwen3-VL-30B-A3B-Instruct improves +24 pp (18 → 42). Within the family the gain grows with scale, consistent with the reasoning-capacity bottleneck seen on Fara-7B.

Variance. Re-running the headline configuration (Claude + AOI full) across two more seeds yields 82%, 78%, 76% (mean 78.7%, std 3.1 pp), so the reported 82% sits at the upper end of the ∼3 pp decoding-noise band. This spread is orthogonal to the McNemar tests, which compare two configurations on the *same* stochastic runs rather than measuring a single score's variability.

5.3 Per-Family Analysis

Figure 6 maps the AOI gain across families and models (full per-cell scores in Appendix A). Three patterns stand out. First, the *audio* families (Podcast, Phone, Meeting) improve from near-impossible to near-solved. These tasks cannot be done from screenshots at all, so audio perception accounts for nearly the entire gain. Second, the *visual-temporal* families (Screencast, Carousel, Transient) improve consistently. The Dashboard family benefits the least, as periodic values are often already visible in a single screenshot. Finally, GAMES is the one family that regresses across models. Being tightly action-latency bound rather than perception-limited, the added per-step observation overhead pushes the agent past its reaction window. AOI is designed for human-paced computer use, not real-time interaction. Two additional model-level observations emerge: the open-source EvoCUA-32B gains most on audio tasks (weak visual grounding leaves more headroom there), while Dashboard and Games cells stay flat-to-negative for every model. The gains concentrate on families with a genuine perceptual bottleneck and are absent where screenshots already suffice.

5.4 Difficulty Breakdown

The AOI improves performance across every difficulty tier (full breakdown in Appendix A, Table 7). For Claude Sonnet 4.6, success rates rise from 57% to 93% on easy tasks, 38% → 85% on

medium tasks, and 20% → 67% on hard tasks. Hard tasks remain the most challenging because they need both strong perception *and* reasoning. On the smallest model (Fara-7B), gains on medium/hard tasks stay small, while gains on easy tasks nearly double. This highlights that Fara-7B's performance ceiling is primarily limited by reasoning capacity rather than perception.

5.5 Efficiency

Better perception leads to fewer but better-aimed steps, resulting in a counter-intuitive cost consequence: *the AOI is cheaper than the screenshot baseline*. Despite its per-step observation overhead, token costs drop by 15–50% across all cloud models (Claude and Gemini −50%, GPT-5.4 −15%). The reduction in step count (roughly 50%) outweighs the richer per-step input. Wall-clock time shows the opposite trend: it *rises* for fast-inference models, where observation work exceeds the latency saved by fewer steps, and falls only for the slowest model, where step reduction dominates. AOI thus lowers cost for batch or cost-sensitive workloads. Latency-sensitive use is a per-model judgement call (full instrumented breakdown in Appendix A, Table 8).

6 Decomposing the Gain

Where does the AOI gain come from? We address this question in two stages. First, we separate the contributions from AOI's prompt format from genuine perception contributions (Section 6.1). Second, we decompose the perception gains channel by channel through targeted ablations (Sections 6.2–6.4).

6.1 Prompt Format vs. Perception

Before attributing the gains to improved perception, we rule out the most concerning confounding factor: that the AOI helps only by reformatting the prompt. Two targeted checks confirm this is not the case.

First, on the **Static-50** benchmark (50 silent HTML tasks such as forms, tables, and calculations whose rendered page never changes), the AOI's gates almost never activate (7 keyframes across all 50 tasks, no audio). Nevertheless, AOI scores a perfect 50/50 compared to the screenshot baseline's 43/50. This perfect score on a workload with near-zero perception extraction demonstrates a genuine prompt-format benefit and confirms that AOI does not degrade performance on static tasks (Appendix B).

Second, on the dynamic benchmark we separate the two components directly. A control condition that supplies the AOI's structured observation record but disables all keyframe and audio extraction isolates the prompt-format contribution. The remainder can be attributed to perception gains. Figure 7 shows that both components are positive for every model, ruling out both the "purely reformatting" and "purely perception" interpretations. The balance is model-specific: Claude is perception-leaning, Gemini 2.5 is prompt-leaning (for a video-native model, much of the gain is simply a better way to present context it already had), and the small Fara-7B cannot exploit the richer prompt at all. Within the prompt-format component, the DOM-element list is the single largest lever (Appendix B). The rest of this section examines the *perception* component in greater detail: what each channel (keyframe selection, keyframe images, audio, narration) contributes, and how.

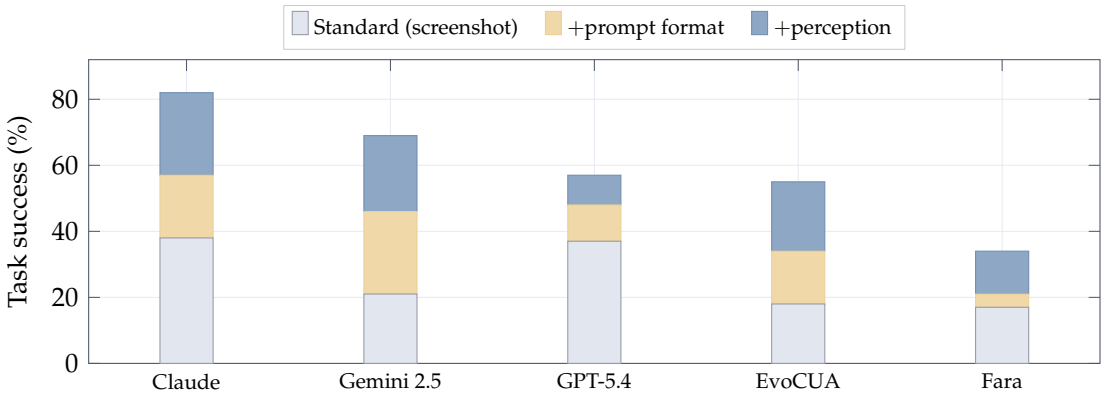


Figure 7. The AOI gain splits into a prompt-format and a perception component—both positive for every model. Bars stack the raw-screenshot baseline, the gain from the AOI's structured observation record alone (no keyframes/audio), and the further gain from perception (keyframes + audio + narration). The worst-case view that the gain is "just reformatting" is ruled out; the split is model-specific (Claude perception-leaning, Gemini 2.5 prompt-leaning, the small Fara-7B unable to exploit a richer prompt). Per-model numbers in Appendix B.

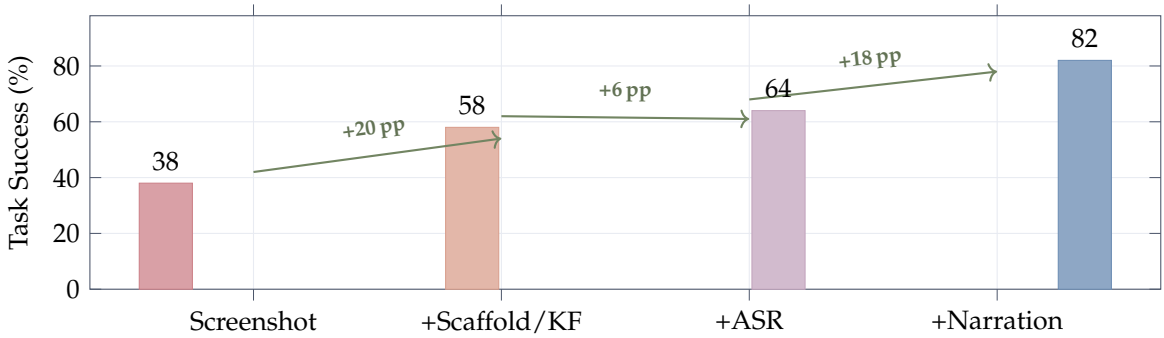


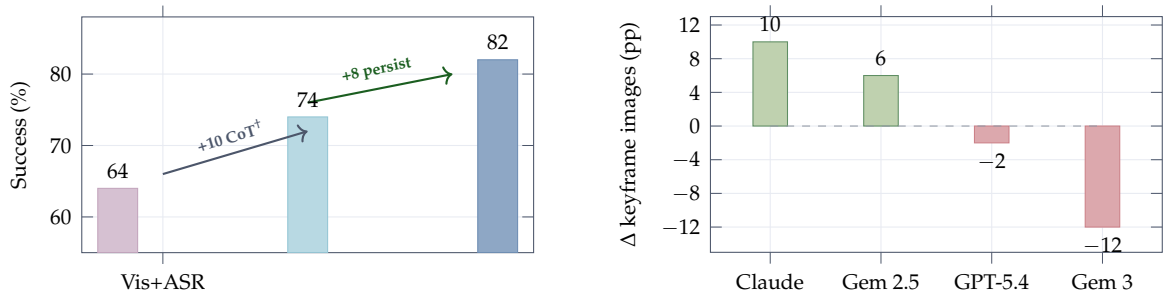
Figure 8. Progressive contribution of each AOI component (Claude Sonnet 4.6). The first step (+20 pp) *bundles* inter-step keyframes with the structured prompt scaffold that bare screenshots lack. The scaffold carries +19 of it (Section 6.1), the keyframe images +1 pp as raw input but +10 pp once narration is present (Figure 9b). ASR adds +6 pp (directional, $p = 0.18$). Visual narration adds +18 pp, the largest single content component. Per-tier significance tests are in Appendix C.

6.2 Keyframe Selection Is Immaterial, and Images Pay Off Through Narration

Figure 8 walks Claude up the component ladder. Two facts about the keyframe channel stand out.

Selection is immaterial. Five visual-only selection strategies (uniform 1 and 3 FPS, pixel-difference, random, and the two-stage CLIP filter of Algorithm 1) all land in a 56–58% band and are statistically indistinguishable (every pairwise McNemar $p > 0.5$, Appendix C). Any inter-step frame breaks the agent's blindness, and *how* it is chosen does not matter. This result also holds on an open-source model (Qwen3-VL-32B, $p > 0.4$ for every pair, Appendix C) and is insensitive to the CLIP threshold across a 15× range (Appendix G). No learned or semantic frame selector is needed.

The images pay off only once narrated. The initial +20 pp gain from adding keyframes is driven almost entirely by the structured observation scaffold that bare screenshots lack (+19 pp on its own, Section 6.1). Holding the scaffold fixed, the raw keyframe images themselves contribute



(a) Narration mechanism (Claude). **(b)** Keyframe-image value, in context.

Figure 9. What narration and keyframes contribute. **(a)** Visual narration lifts Claude from 64% (visual+ASR, no narration) to 82%: discarding the narration text after it is generated keeps 74%, splitting the +18 pp into ∼+10 pp inference-time articulation (†directional, p=0.12) and +8 pp from persisting it as memory (p=0.039). **(b)** The marginal value of the inter-step keyframe *images*, measured with audio and narration already present, is model-specific: +10/+6 pp on Claude/Gemini 2.5 (realised only through narration—just +1 pp as raw input), neutral on GPT-5.4, and a −12 pp *penalty* on Gemini 3 (image-token dilution, Section 7).

only +1 pp. However, once the model narrates what it sees, the same keyframes increase their worth to +10 pp (Figure 9b). This lift occurs precisely on the task families where content changes *between* steps (Transient, Carousel), rather than where one screenshot per step already suffices. The keyframe channel is thus synergistic with narration. Its model-dependence is the subject of Section 6.4.

Audio and narration. Speech transcription improves performance from 58% to 64% (directional at N=100, $p = 0.18$, and confirmed at scale by the audio-family gains of Figure 6 and the Gemini 3 decomposition of Section 7), with gains concentrated on the Phone and Interview families. Visual narration is the largest single content component, yielding an +18 pp improvement (64% → 82%, $p = 5.3 \times 10^{-4}$). Whether this benefit arises due to persistent memory or as inference-time articulation is the question of the next subsection.

6.3 Narration: Articulation or Persistent Memory?

Narration does two things at once: it makes the model *articulate* the new visual content during reasoning, and it *persists* that content as text that survives keyframe pruning. To separate these effects, we run a *narration-discarded* control: the model narrates exactly as in AOI full (preserving any inference-time benefit), but the resulting narration text is dropped before it can be reused. This condition achieves 74/100, between visual+ASR (64) and full (82) (Figure 9a), splitting the +18 pp gain into a directional +10 pp from articulation (64 → 74, $p = 0.12$, underpowered since $p < 0.05$ would need $N \sim 300$) and a significant +8 pp from persistence (74 → 82, $p = 0.039$).

What is the persistence effect doing? An external-judge audit of the ten tasks that full AOI wins but narration-discarded loses finds the prior-step narrations contained the load-bearing fact in only two cases, both multi-step tasks (breakpoints recorded across video frames and page titles recorded as a sequence advanced). In the other eight, the necessary information was present *within* the winning step itself. In those cases, persistence is not supplying a verbatim answer, but steering earlier steps onto a better trajectory (three first-step wins are pure decoding variance). Narration's primary mechanism is therefore articulation. Persistence

Table 3. Later-released models (100 dynamic tasks each, same harness). Gemini 3 checks whether the Gemini 2.5 +48 pp gain replicates. The two Grok rows control for the inference-latency floor.

Model	Standard	AOI full	Δ (pp)	p (McNemar)
Gemini 3 Flash	36	45	+9	0.18
Grok-4.3	25	65	+40	8.2×10^{-10}
Grok-4-fast-reasoning	19	47	+28	4.9×10^{-6}

matters most where information genuinely spans steps.

6.4 The Keyframe Channel Is the Most Model-Dependent

Measured in the deployed configuration (audio and narration already on), the marginal value of the inter-step keyframe *images* swings from +10 pp on Claude and +6 pp on Gemini 2.5 (both realised only through narration, since the same images add +1 pp without it), through a neutral −2 pp on GPT-5.4, to a −12 pp *penalty* on Gemini 3 (Figure 9b). The positive gains concentrates on the families where frames carry novel between-step content (Transient, Carousel) and is near-zero on audio and static families, pinning it to narration rather than audio. The observation generalises into the operative rule of Section 7: information delivered as *text* (transcripts and narration) is robustly positive on every model, whereas the raw image stream must be treated as a per-model toggle. The gates make that toggle cheap: only ∼28% of steps ever produce a keyframe and ∼14% activate audio, leaving ∼61% of steps fully idle (per-family activity in Appendix D). A cross-engine check (audio re-rendered with a different speech synthesizer and real terminal-recording screencasts) confirms the audio pipeline does not silently fail under an unfamiliar acoustic profile (Appendix J).

7 Model Case Studies: Later-Released Models

Do the AOI's gains hold up on models released after the main runs? We run three additional 100-task evaluations with the same harness (Table 3). Together with the original Grok-4 results, they form two instructive case studies: a family where the binding constraint is inference *latency* rather than perception (Grok), and a model where one AOI component *reduces* performance (Gemini 3).

Grok: latency, not perception, is the ceiling. The original Grok-4 scored only 4/100 in the standard setting and 25 with AOI (+21 pp, $p = 3 \times 10^{-6}$): actions are valid, but a single model call takes 80–180 s, causing the agent to time out after one or two steps on harder tasks. The AOI helps where a few well-placed observations suffice, but cannot unblock tasks needing many sequential actions, so +21 pp lower-bounds the true effect. Follow-up evaluations confirm this: a lower-latency variant reaches 47, and the newer Grok-4.3 reaches 65, decomposing the ceiling into ∼+22 pp from lowering latency and ∼+18 pp from base-model improvement.

Gemini 3: a component reduces performance. Gemini 3 Flash nets only +9 pp ($p = 0.18$), down from Gemini 2.5's +48. This does not reflect the model having internalised the AOI: a four-way decomposition (Figure 10, with tables in Appendix E) shows it is a *cancellation* of effects. Audio still helps (+12 pp, with +10 on the audio families alone, so Gemini 3 is not audio-saturated), and the structured scaffold still helps on balance (+9 pp), but the inter-step keyframe images now *regress* −12 pp. Dropping just the keyframes recovers 57/100, a +21 pp

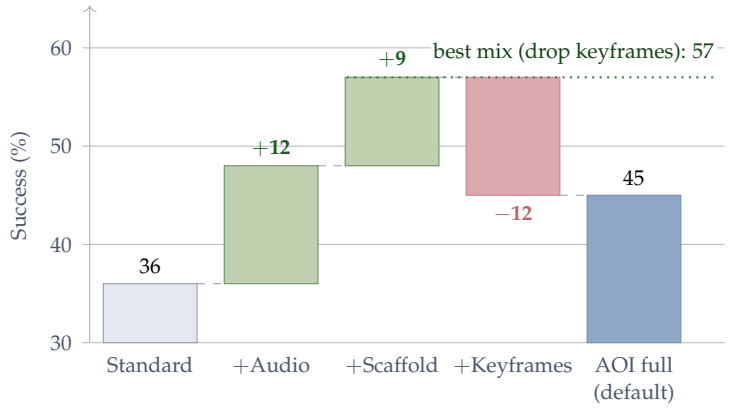


Figure 10. The AOI's small +9 pp net on Gemini 3 Flash is a three-way cancellation, not saturation. Audio (+12) and the structured scaffold (+9) help as on every other model, but the keyframe-image stream *regresses* −12 pp—so the default bundle nets only 36 → 45. Dropping just the keyframes recovers 57/100 (+21 pp over Standard), showing the slack is real and the bundle merely mis-tuned. Underlying per-mode and per-family tables are in Appendix E.

gain. The slack remains and the default bundle is simply miscalibrated. A causal probe (Appendix E) isolates the mechanism as *image-token dilution*, where extra images degrade Gemini 3's action grounding regardless of content, rather than distraction by what the frames show. The operating rule for this model is simple: deliver observation as text and turn the keyframe-image stream off.

A diagnostic map of model slack. Read this way, the AOI's residual gain is a diagnostic of *where* a model has headroom, along richer axes than perception alone: prompt-handling (Gemini 2.5 leans prompt-format, +25/ + 23), perception (Claude leans perception, +19/ + 25), reasoning capacity (Fara-7B cannot exploit a richer prompt, +4/ + 13), latency (Grok), and action-policy compatibility with the scaffold (Gemini 3). The interface is not a fixed bundle. Its components must be selected per model.

8 Comparison with Streaming Multimodal Baselines

A natural alternative is a *streaming, end-to-end multimodal model* that bundles perception and reasoning in one trained model. Today, these are Gemini Live [Google, 2025] and the OpenAI Realtime API [OpenAI, 2024]. Neither ships as a CU agent. Each is a voice-first assistant with function-calling. We adapt them to DynaCU-Bench (one session per task, the page's native audio and screenshots streamed in over the live websocket, returned tool calls executed in the browser) and score them on the same 12-task audio subset the AOI is scored on. To pre-empt the objection that a stronger model would close the gap, we evaluate the current generation directly: OpenAI's GA Realtime model (`gpt-realtime-2`), streaming the page's native audio rather than a transcript. We run it in two conditions: *alone* (screenshot + audio, the information a streaming model has on its own) and + *AOI scaffold* (additionally handed the same `[PAGE ELEMENTS]` interactive-element list the AOI provides), which separates a perception deficit from an action-grounding one. We also evaluate xAI's newly released Grok Voice Agent (`grok-voice-think-fast`), which speaks the same GA Realtime protocol but accepts *audio and text only* (no vision), so it runs with native audio plus the text scaffold and never sees the screen.

Table 4. Streaming multimodal baselines vs. the AOI on a 12-task audio subset (3 each from Podcast, Meeting, Phone, Interview). Each row is scored on the same task IDs as the AOI in Table 2. `gpt-realtime-2` is the current GA OpenAI Realtime model, streaming native audio. "alone" is screenshot + audio, and "+ scaffold" additionally supplies AOI's `[PAGE ELEMENTS]` list. Grok Voice is audio+text only (no vision), shown in both conditions. This is a deliberately favourable comparison, as no baseline is built for grounded computer use.

Configuration	Pod	Meet	Phone	Intv	Total / 12
Gemini Live (2.5)	0	0	0	0	0 (0%)
OpenAI Realtime (gpt-4o)	0	0	0	3	3 (25%)
Grok Voice (no vision), audio only	1	0	0	0	1 (8%)
Grok Voice (no vision), + scaffold	1	0	0	0	1 (8%)
gpt-realtime-2, alone	0	0	1	1	2 (17%)
gpt-realtime-2, + AOI scaffold	2	3	3	3	11 (92%)
AOI full (Claude 4.6)	3	3	3	3	**12** (100%)

A streaming voice model does not substitute for the AOI, and the current generation sharpens *why* (Table 4). The older baselines fail outright: Gemini Live returns few tool calls and clicks non-actionable coordinates, and the gpt-4o-backed Realtime succeeds only on the simplest interview questions. Crucially, `gpt-realtime-2`, the current GA Realtime model with native audio, still fails on its own, even though its transcription is essentially perfect (the run logs show it hears every name and figure correctly). Its failure is not perception but *action grounding*: it hears "Dr. Sarah Chen" yet cannot target the right form field from a screenshot, so the answer never lands. Handed AOI's `[PAGE ELEMENTS]` scaffold, the very same model recovers most of the gap, which localises the deficit precisely: a frontier streaming model already perceives the page, but needs the AOI's structured grounding to act on it, and even then only ties the AOI. xAI's newly released Grok Voice (`grok-voice-think-fast`) makes the same point from the opposite direction. It hears accurately (on one task it transcribes a spoken price and types it correctly), but, lacking vision entirely, it fails whether or not it is given the text scaffold, idling on most steps. Unlike `gpt-realtime-2`, it does not exploit the element list, defaulting to blind clicks and occasional hallucinated answers. Perception without grounded action is not enough. A sanity check rules out a broken adapter: on a set of purely-visual tasks the older baselines emit valid actions on every task but get none task-correct, while `gpt-realtime-2` trails the AOI-equipped Claude reference by one task (Appendix K).

9 Related Work

Expanding the interface, not the model. Much agent progress comes from reshaping the agent–computer interface rather than the weights. On the *action* side, SWE-agent [Yang et al., 2024] redesigned the command set and feedback format, and agents adopt executable code actions [Wang et al., 2024a] and element-grounded actions [Deng et al., 2023]. The *observation* side is also well studied, but almost entirely as richer encodings of a *single static snapshot*: accessibility trees and HTML/DOM text surrogates [Zhou et al., 2023, Deng et al., 2023] (which serve as both observation and action representations), and set-of-mark element overlays that ground vision models on the screenshot [Yang et al., 2023, Zheng et al., 2024]. What none of these representations change is *when* the agent looks: one snapshot per action, and no audio. We expand observation along the orthogonal axes they leave out, time (continuous,

between-step capture) and modality (sound). This is complementary to single-frame encodings and composes with them.

Computer-use agents. The CU agent landscape spans closed-source systems [Anthropic, 2024a, OpenAI, 2025, Google DeepMind, 2025, Andreux et al., 2025] and open-source models [Wang et al., 2025a,b, Awadallah et al., 2025, Ye et al., 2025, Song et al., 2025]. Agent S3 [Simular AI, 2025] demonstrates competitive open-source performance on OSWorld. A recent CU agent survey [Sager et al., 2025] catalogues many agents and datasets and identifies several open gaps, but does not identify the observation modality as one. All of these agents operate through the screenshot–reason–act loop. None addresses dynamic visual content or audio perception.

Real-time multimodal models. Gemini Live [Google, 2025] and the OpenAI Realtime API [OpenAI, 2024] process continuous audio/video/text streams in real time, bundling perception with reasoning into a single trained model. Our work is complementary: the AOI decouples perception from reasoning, so the same perception layer benefits any CU model without retraining. Section 8 compares the two paradigms head-to-head on the audio-focused DynaCU-Bench subset.

Video understanding for GUI. GUI-World [Chen et al., 2024], VideoWebArena [Jang et al., 2024], and CUA-Suite [Chen et al., 2025] identify the observation problem through benchmarking. MemGUI-Bench [Park et al., 2025] reveals memory gaps in GUI agents. These benchmarks identify the problem but do not propose solutions compatible with existing CU agents.

Keyframe selection. VideoTree [Wang et al., 2024b] and AKS [Tang et al., 2025] use CLIP-based frame selection for offline video QA. Apollo [Zohar et al., 2024] finds that FPS sampling outperforms uniform sampling and identifies optimal token budgets. All prior work targets offline video understanding. Our contribution is real-time observation filtering for CU agents, where the two-stage design (sub-millisecond pixel gate followed by $\sim$7 ms CLIP only on changed frames) keeps the per-frame cost low enough to run continuously in the background between agent steps.

Adaptive middleware for GUI agents. Cradle [Tan et al., 2024] includes frame difference analysis for games, the closest predecessor to the AOI's perception layer, but uses pixel-level differencing without semantic filtering and has no audio processing. ScreenLLM [Jin et al., 2025] captures stateful screen schemas with keyframe extraction, complementing visual narration but without audio or adaptive observation. StreamAgent [Yang et al., 2025] and Dispider [Qian et al., 2025] separate perception from response generation for streaming video, paralleling the AOI's architecture, but neither addresses CU agent observation.

Structured agent interfaces. A complementary line of work bypasses the perception problem by having websites or applications expose structured interfaces directly to agents: the Model Context Protocol [Anthropic, 2024b] for tool/data exchange and declarative web specifications such as NLWeb [Microsoft, 2025] for site-side agent endpoints. These approaches are powerful where adoption exists, but require website cooperation. The AOI enables perception on arbitrary, unmodified web content (any HTML/audio rendered to a browser, including legacy or third-party pages with no agent endpoint).

10 Limitations

Latency and temporal resolution. The AOI extends perception but not decision speed: CU models still take 1–5 s per step, and the screen is sampled at ∼3 Hz, so sub-∼300 ms events and real-time games fall outside its reach (the Games family regression). It targets human-paced computer use, not fast interactive control.

Perception is not reasoning. Better inputs do not improve reasoning over those inputs. On the smallest model the gain concentrates on easy tasks and reasoning capacity becomes the binding constraint.

Audio is speech-only and narration can err. We transcribe speech but not non-speech events (chimes, beeps). A multimodal audio model would be necessary to cover the full audio scene. Long-term visual memory also inherits the CU model's mistakes: a misread number persists in the trajectory.

External validity. Tasks run in a controlled browser with synthesized audio, so real-world content (live calls, native apps) may differ, and we do not yet report AOI numbers on existing dynamic-GUI benchmarks. Those evaluate offline video QA or fixed-context video rather than a live observe–act loop with a separate audio channel, so wrapping them with the AOI requires re-instrumenting each harness. That re-instrumentation is the clearest external-validity test and the main item of future work.

Statistical power. Each cell is a single 100-task trial. We use paired McNemar tests for differences, but the data does not support fine-grained ranking of configurations within a few points of each other.

11 Conclusion

Computer-use agents are blind between screenshots and deaf to audio because observation is tied to action: one screenshot per step. We argued the *observation* interface is a separable, underexplored design axis, the counterpart to the action interface SWE-agent surfaced. We introduced the AOI, a perception layer that is transparent on static work, adaptive on dynamic content, and compatible with any existing CU model without retraining. Across eight models it turns dynamic tasks that were near-impossible from screenshots into largely solvable ones, while reducing token cost.

Opening up that gain reframes CU perception in three ways. First, keyframe *selection* is immaterial and value emerges only once frames are narrated into persistent text. Second, narration helps mainly by making the model articulate what it sees. Third, the interface is not a fixed bundle. One component even reverses sign on a later-generation model, so the components must be tuned per model rather than shipped as one configuration. We release the perception layer and benchmarks so future work can cleanly separate observation gains from model-side gains.

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A Detailed Results

This appendix provides the per-cell numbers summarised by the figures in the main paper.

- Per-family success for every model (Table 6, the absolute scores behind the gain heatmap in Figure 6).
- The difficulty breakdown (Table 7).
- Efficiency and cost (Table 8).
- The full per-family ablation sweep (Table 9).
- The additional open-source replication on the Qwen3-VL family (Table 5).

Table 5. Additional open-source replication on the full 100-task DynaCU-Bench: Qwen3-VL family via OpenRouter (provider: DeepInfra), evaluated with the same harness as the six models in Table 2. p-values are paired McNemar exact tests on the discordant tasks (b/c = success only under standard / only under AOI). Both gains fall inside the +17 to +48 pp band of the main table. Two further candidates (UI-TARS-1.5-7B, GLM-4.5V) were unavailable through our provider list at evaluation time.

Model	Standard	AOI full	Δ (pp)	b/c	p (McNemar)
Qwen3-VL-235B-A22B-Instruct	22	**64**	+42	4/46	4.5×10^{-10}
Qwen3-VL-30B-A3B-Instruct	18	**42**	+24	8/32	1.8×10^{-4}

Table 6. Per-family success rates (passes out of 10) for Standard vs. AOI full across the five models with full per-family breakdowns. Grok-4 is omitted due to latency constraints (see Section 7). These are the absolute scores behind Figure 6.

	Claude 4.6		GPT-5.4		Gemini 2.5		EvoCUA-32B		Fara-7B	
Family	Std	AOI	Std	AOI	Std	AOI	Std	AOI	Std	AOI
Podcast	0	**9**	0	3	0	**8**	0	**8**	0	3
Meeting	2	**10**	2	3	2	**8**	1	6	1	5
Screencast	4	**9**	5	**8**	5	6	3	6	3	5
Carousel	4	**9**	6	7	4	7	3	4	1	3
Dashboard	8	8	6	**9**	1	**9**	2	4	2	2
Transient	4	**9**	2	**8**	0	**9**	0	5	4	3
Phone	1	**9**	1	4	0	**8**	0	**8**	1	4
Interview	7	**9**	6	7	5	7	6	**8**	1	3
Collab	3	**8**	4	5	2	6	2	5	2	4
Games	5	2	5	3	2	1	1	1	2	2
Total	38	**82**	37	**57**	21	**69**	18	**55**	17	**34**

B Prompt-Format vs. Perception Decomposition

These are the numbers behind Figure 7.

- Table 10: The Static-50 no-degradation check.
- Table 11: The per-model prompt/perception split.
- Table 12: Fine-grained breakdown of the prompt-format effect, showing that a DOM-element list is the single largest contributor.

Table 7. Task success by difficulty (Standard → AOI full). Easy: 30 tasks. Medium: 40. Hard: 30.

	Standard			AOI Full		
Model	Easy	Med	Hard	Easy	Med	Hard
Claude 4.6	17/30	15/40	6/30	**28**/30	**34**/40	**20**/30
GPT-5.4	15/30	17/40	5/30	21/30	26/40	10/30
Gemini 2.5	10/30	9/40	2/30	24/30	33/40	12/30
EvoCUA-32B	6/30	8/40	4/30	20/30	26/40	9/30
Fara-7B	9/30	5/40	3/30	19/30	11/40	4/30

Table 8. Efficiency and cost on the 100-task DynaCU-Bench. Grok-4 is omitted due to inference latency constraints that distort the comparison. Steps = mean steps per task. Time = mean wall-clock per task. Lat. = total inference latency per task. Obs = total observation processing time per task. Tokens = input + output tokens per task (estimated, same estimator for every row). $/100 = cost per 100 tasks at May 2026 list prices (local models free). The AOI reduces token cost on every cloud model because the reduction step count more than offsets the increased input size.

Configuration	Steps	Time (s)	Lat. (s)	Obs (s)	Tokens (k)	$/100
Claude Sonnet 4.6						
Standard	10.7	40.6	26.3	2.5	7.6	$2.72
AOI full	**4.8**	46.2	15.2	12.0	**3.8**	**$1.35**
GPT-5.4						
Standard	10.0	26.6	14.4	2.4	5.8	$3.23
AOI full	**7.6**	50.6	11.2	17.8	**5.0**	**$2.76**
Gemini 2.5 Flash						
Standard	11.5	55.3	40.4	2.6	7.6	$0.32
AOI full	**5.4**	61.9	20.6	20.6	**4.0**	**$0.16**
EvoCUA-32B (local)						
Standard	9.3	117.0	99.0	2.3	5.8	—
AOI full	**5.3**	113.9	72.0	22.6	**4.0**	—
Fara-7B (local)						
Standard	13.4	35.1	24.0	2.5	9.2	—
AOI full	**9.3**	90.7	17.0	31.6	**7.1**	—

C Statistical Tests and Component Decompositions

This appendix includes:

- The per-tier significance tests behind Figure 8 (Table 13).
- The open-source replication of the selection-invariance finding (Table 14).
- The per-family narration decomposition behind Figure 9a (Table 15).
- The in-context keyframe-image decomposition behind Figure 9b (Table 16).

D Per-Step Observation Activity

The gates suppress processing on the majority of steps: only ∼28% produce a keyframe and ∼14% activate audio (Table 17), and weighted by step count, ∼61% of steps are fully idle (Figure 12). Audio-only families produce zero keyframes and visual-only families produce

Table 9. Complete per-family results for all 8 observation configurations tested with Claude Sonnet 4.6. Family abbreviations: Pod=Podcast, Meet=Meeting, Scr=Screencast, Car=Carousel, Dash=Dashboard, Tran=Transient, Phn=Phone, Int=Interview, Col=Collab, Gam=Games.

Mode	Pod	Meet	Scr	Car	Dash	Tran	Phn	Int	Col	Gam	**Total**
Standard	0	2	4	4	8	4	1	7	3	5	38
Uniform 1 FPS	0	4	10	8	9	8	1	8	6	4	58
Uniform 3 FPS	2	5	8	8	8	8	1	7	6	3	56
Pixel-diff	1	5	9	7	9	8	1	8	6	4	58
Random keyframes	1	6	8	6	9	8	1	8	6	3	56
AOI visual	1	5	9	7	8	9	1	8	6	4	58
AOI visual+ASR	1	4	10	9	9	9	4	9	6	3	64
AOI full	**9**	**10**	**9**	**9**	**8**	**9**	**9**	**9**	**8**	2	**82**

Table 10. Static-50 (50 purely static, silent tasks) on Claude Sonnet 4.6. The gates suppress almost all extraction (7 keyframes total, no audio), yet the AOI scores 50/50. This shows the presence of a prompt-format effect, not improved perception, as well as no degradation on static work.

Mode	**Pass/50**	**Avg Steps**	**Total KF**	**Audio Seg.**
Standard	43 (86%)	3.26	0	0
AOI full	**50** (100%)	**1.62**	7	0

Table 11. Prompt-format vs. perception decomposition on DynaCU-Bench (100 tasks). A structured-record control with no keyframes/audio isolates the prompt format gain Δ_{prompt}. The remainder up to AOI full is pure perception gain Δ_{percept}. Both components are positive for every model. (Results for Gemini 3 Flash, where the split differs, is in Appendix E.)

Model	**Standard**	**+format**	**AOI full**	$\Delta_{\textbf{prompt}}$	$\Delta_{\textbf{percept}}$
Claude Sonnet 4.6	38	57	82	+19	+25
GPT-5.4	37	48	57	+11	+9
Gemini 2.5 Flash	21	46	69	+25	+23
EvoCUA-32B	18	34	55	+16	+21
Fara-7B	17	21	34	+4	+13

Table 12. Which prompt sub-component carries the load (Claude Sonnet 4.6, 100 tasks). Trajectory = prior-step action/text history and sectioning. PAGE EL = DOM-element list. The element list is the single largest lever (+30 pp added to the minimal prompt), and the two are sub-additive.

Mode	**Pass/100**	**Trajectory?**	**Element list?**
Minimal (task + screenshot)	16	no	no
+trajectory (=Standard)	38	yes	no
+element list	46	no	yes
+both (=+format)	**57**	yes	yes

zero audio activations, confirming clean channel separation. Figure 11 traces this step-by-step on two representative tasks.

Table 13. Pairwise McNemar exact mid-p tests between successive ablation tiers (Claude Sonnet 4.6, 100 tasks). b/c are discordant counts (success only in the previous / only the current configuration). The selection-method group is statistically indistinguishable. The between-tier transitions are significant.

Comparison	Total	b	c	p
Standard → Uniform 1 FPS	38 / 58	3	23	8.8×10^{-5}
Uniform 1 FPS → Uniform 3 FPS	58 / 56	7	5	0.77
Uniform 3 FPS → Pixel-diff	56 / 58	4	6	0.75
Pixel-diff → Random keyframes	58 / 56	4	2	0.69
Random keyframes → CLIP adaptive	56 / 58	4	6	0.75
CLIP adaptive → AOI visual+ASR	58 / 64	4	10	0.18
AOI visual+ASR → AOI full	64 / 82	4	22	5.3×10^{-4}

Table 14. Selection-method ablation replicated on an open-source model (Qwen3-VL-32B), 50 visual-temporal tasks (Screencast, Carousel, Dashboard, Transient, Games). The three methods cluster with no pairwise significance ($p > 0.4$), mirroring Claude: the convergence is not model-specific.

Selection method	Pass / 50	Vs. adjacent
Uniform 1 FPS	26	—
Pixel-diff	27	vs. Uniform: $p = 0.69$
Random keyframes	27	vs. Pixel-diff: $p = 1.00$

Table 15. Narration decomposition (Claude Sonnet 4.6, 100 tasks): visual+ASR (no narration) → narration-discarded → full. The discarded run isolates the inference-time articulation effect, and the gap to full isolates persistence. Per-family rows are descriptive (single-trial). Note Transient reads 9→4→9, a decoding fluctuation on a 10-task cell that the totals average out.

Configuration	Tot.	Pod	Meet	Scr	Car	Dash	Tran	Phn	Int	Col	Gam	Vs. adj.
Visual+ASR (no narr.)	64	1	4	10	9	9	9	4	9	6	3	—
Full, narration discarded	74	9	10	8	8	7	4	9	9	7	3	$p = 0.12$
Full (narration persisted)	**82**	9	10	9	9	8	**9**	9	9	8	2	$p = 0.039$

Table 16. Marginal value of the inter-step keyframe *images* measured in context (audio and narration present): scaffold+ASR+narration without keyframes vs. with them. Δ_{KF} ranges from +10 pp to a −12 pp penalty across models, the most model-dependent component. †GPT-5.4 here is measured through a single backend for both modes, so only its within-row Δ_{KF} is comparable to the absolute scores elsewhere.

Model	no keyframes	+keyframes	Δ_{KF} (pp)
Claude Sonnet 4.6	72	**82**	+10
Gemini 2.5 Flash	63	**69**	+6
GPT-5.4†	**72**	70	−2
Gemini 3 Flash	**57**	45	−12

E Gemini-3 Flash Decomposition

This appendix provides the tables underlying the Gemini 3 case study (Figure 10).

- Table 18 is the four-way decomposition behind the waterfall.
- Table 19 is the per-family AOI Δ (the shortfall vs. Gemini 2.5 sits entirely on Dashboard and

Meeting task — "What is the new launch date?"

Standard (screenshot-only) — FAIL

15 steps, 51.2 s

1. `wait()` — sees static slide
2. `wait()` — same slide
3. `wait()` — same slide
4. `wait()` — same slide
5. `wait()` — same slide

⋮

15. `wait()` × step budget exhausted

Agent never hears the spoken date.
Waits all 15 steps without ever acting.

AOI (full) — PASS

2 steps, 26.2 s

1. `wait()`
   Audio: "Here's the team. As you can see we have strong cross-functional coverage."
   Narration: "Meeting slide shows Project Team with 6 members."
2. `fill("April 28th")` ✓
   Audio: "We've moved the launch date to April 28th. Please update."
   Narration: "Audio announced the new launch date."

Agent hears "April 28th" and fills the form immediately.

Screencast task — "What was the first command in the terminal recording?"

Standard — PASS (6 steps)

1. `wait()` — recording playing
2. `wait()` — missed first command
3. `wait()` — scrolling through
4. `wait()`
5. `wait()`
6. `fill(...)` ✓ lucky timing

AOI — PASS (1 step)

1. `fill("git clone ...")` ✓
   Narration: "The terminal recording shows the first command was git clone…"

Narration captures the command from the recording immediately.

Figure 11. Trajectory comparison on two representative tasks (*illustrative, cherry-picked to expose the mechanism*). **Top:** Meeting task where the launch date is spoken aloud. The standard agent cannot hear it and exhausts its step budget on 15 `wait()` steps. The AOI agent transcribes the audio, hears "April 28th," and acts in 2 steps. **Bottom:** Screencast task where the answer appears in a terminal recording. The standard agent needs 6 steps with lucky timing. The AOI agent's narration captures the content in 1 step.

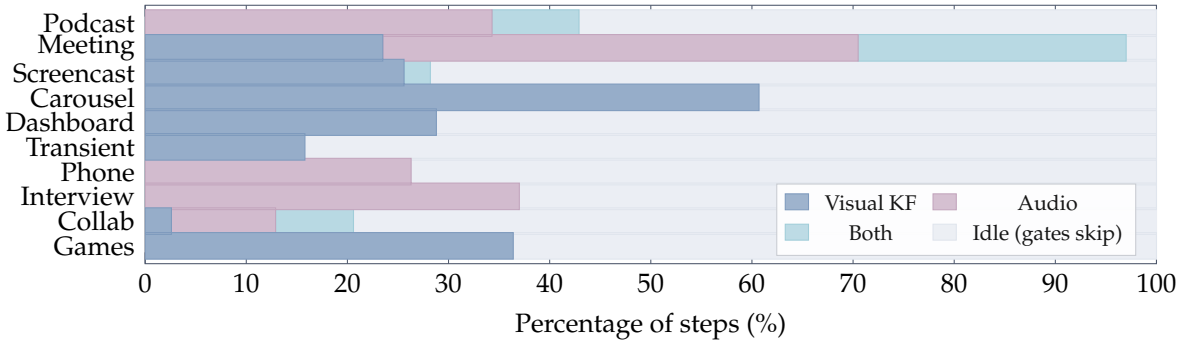

Figure 12. Observation activity per family. Percentage of steps in which each channel fires, and gray is idle (both gates suppress). Carousel is highly visual. Phone and Interview are audio-only. Meeting uses both. ∼61% of steps are idle overall.

Table 17. Per-step observation statistics for Claude Sonnet 4.6 + AOI full, by family. %KF = steps capturing ≥ 1 keyframe. KF/Step = mean keyframes/step. %Audio = steps with audio transcription.

Family	Steps	% KF	Tot. KF	KF/Step	% Audio
Podcast	35	8.6	3	0.09	42.9
Meeting	34	50.0	18	0.53	73.5
Screencast	39	28.2	12	0.31	2.6
Carousel	61	60.7	84	1.38	0.0
Dashboard	52	28.8	18	0.35	0.0
Transient	38	15.8	7	0.18	0.0
Phone	38	0.0	0	0.00	26.3
Interview	27	0.0	0	0.00	37.0
Collab	39	10.3	4	0.10	17.9
Games	118	36.4	65	0.55	0.0
All	**481**	**28.3**	**211**	**0.44**	**14.1**

Transient, where Gemini 3 alone regresses).

- Table 20 is the causal probe that pins the keyframe regression to image-token dilution.

Table 18. Four-way decomposition of the Gemini 3 AOI residual. If the small net were internal saturation, the audio-plus-standard-prompt row would not exceed the full AOI. "Aud4" = the four audio families (Podcast, Meeting, Phone, Interview). "D+T" = Dashboard + Transient, where Gemini 3 regresses.

Mode	Total	Aud4	D+T	Scaffold?	Audio?	Keyframes?
Standard	36	10	11	no	no	no
Standard+scaffold	29	9	2	yes	no	no
Standard+audio	48	20	7	no	yes	no
Scaffold+audio (best)	**57**	35	3	yes	yes	no
AOI full (default)	45	26	1	yes	yes	yes

Table 19. Gemini 3 Flash per-family success (pass/10), Standard vs. AOI full. The four audio families gain +16 pp in aggregate. The shortfall vs. Gemini 2.5's +48 pp is concentrated in the Dashboard and Transient regressions.

Mode	Pod	Meet	Scr	Car	Dash	Tran	Phn	Int	Col	Gam	**Total**
Standard	0	2	4	4	**8**	3	1	7	5	2	36
AOI full	**6**	**6**	**6**	**5**	1	0	**6**	**8**	5	2	45
Δ	+6	+4	+2	+1	−7	−3	+5	+1	0	0	+9

F DynaCU-Bench Task List

Each of the 100 tasks is identified by a category letter (A–J) and a difficulty-indexed ID: E1–E3 for easy tasks, M1–M4 for medium tasks, and H1–H3 for hard tasks.

Example tasks by category:

- **A-E1** (Podcast, Easy): Listen to a podcast segment and identify the guest's name.

Table 20. Keyframe causal probe on Gemini 3 (50 visual-active tasks), holding audio and scaffold fixed and varying only the keyframe channel. Replacing keyframes with same-size noise images reproduces the penalty, and the cost does not scale with count or position, pointing to image-token dilution. Conditions are pairwise indistinguishable ($p \geq 0.375$ at N=50).

Mode	Pass / 50	Rate	What it tests
No keyframes (anchor)	**24**	48%	—
Budget 1 keyframe	20	40%	does count matter?
Budget 2 keyframes	18	36%	
Budget 3 keyframes	20	40%	
Default (budget 5)	18	36%	
Noise images	17	34%	content vs. presence
Duplicate-screenshot images	20	40%	content vs. presence
Keyframes after screenshot	22	44%	position effect

- **B-M2** (Meeting, Medium): During a meeting with slides, count the number of items in a presented chart.
- **C-H1** (Video, Hard): Watch a terminal recording and describe the full workflow demonstrated.
- **D-M4** (Carousel, Medium): Identify a specific product name from a rotating carousel.
- **E-E3** (Dashboard, Easy): Read the current warning level from a live-updating dashboard.
- **F-E2** (Transient UI, Easy): Click "Accept" on a cookie consent banner before it auto-dismisses.
- **G-M1** (Phone, Medium): Follow spoken instructions during a simulated phone call.
- **H-H2** (Interview, Hard): Complete a multi-question spoken interview with scoring.
- **I-M2** (Collab, Medium): Resolve a conflict in a collaborative document when a remote edit arrives.
- **J-E2** (Games, Easy): Complete 3 rounds of a simple reaction-time game.

All tasks are implemented as self-contained HTML files with embedded JavaScript for task logic, audio synthesis, and evaluation. The complete benchmark (100 HTML files, evaluation scripts, and task definitions) is released publicly with the code.

G CLIP Threshold Calibration

The default threshold $\theta = 0.04$ was calibrated empirically. Web UI events such as modal dialogs appearing produce cosine distances of $\sim$0.08. Video scene cuts produce distances of $\sim$0.15–0.25. Loading spinners and cursor blinks produce distances of $\sim$0.005–0.02. Setting $\theta = 0.04$ captures all meaningful UI changes while filtering most periodic noise.

Figure 13 shows the CLIP threshold θ sweeps across a 15$\times$ range from 0.02 to 0.30, evaluated on the 40 visual tasks (categories C–F) where keyframe selection is relevant. Point estimates remain within an 80–90% band throughout, demonstrating that for practitioners who choose the CLIP-based variant, threshold selection is not critical. There is no cliff: even at $\theta = 0.30$ (very aggressive filtering that captures only major visual changes), the point estimate remains at 85%. At $\theta = 0.02$ (very sensitive, capturing small changes), it is 82.5%. The point estimate peaks at $\theta = 0.08$ (90%), but with N=40 the 95% Wilson confidence intervals on every point span $\sim$10–12 pp and overlap heavily. We therefore do not claim a true sweet spot, only that the method is robust across a 15$\times$ range of θ and that any reasonable choice will work. This

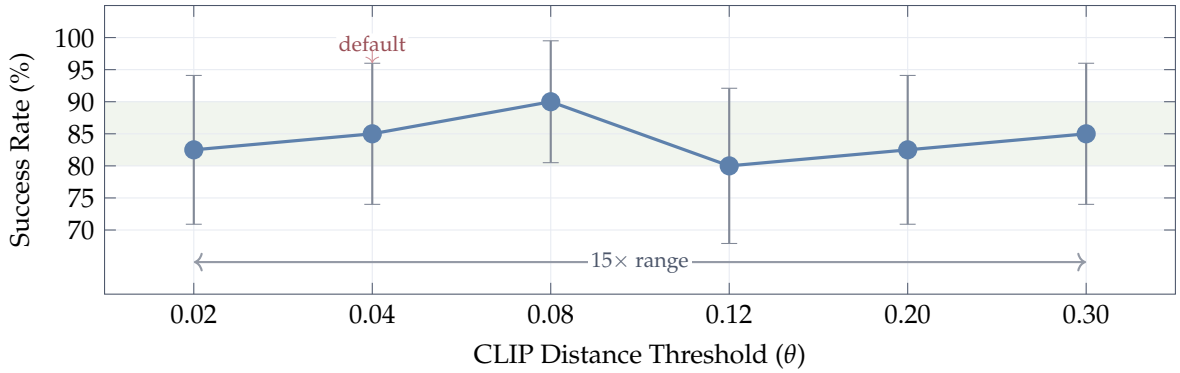

Figure 13. CLIP threshold (θ) sensitivity on 40 visual tasks (categories C–F) with Claude Sonnet 4.6. Error bars are 95% Wilson confidence intervals at N=40 ($\pm$9.5–12.1 pp). Point estimates remain within an 80–90% band (shaded) across a 15$\times$ range of θ, but the CIs overlap heavily, so we cannot distinguish individual thresholds at this sample size. θ values are placed at uniform spacing on the axis (not to scale). The bimodal distribution of CLIP distances (near-zero for unchanged content, large for meaningful changes) makes the method naturally insensitive to θ.

insensitivity arises because CLIP embeddings produce a bimodal distribution of distances: semantically unchanged frames cluster near zero, while meaningful changes produce large distances, with relatively few samples in between.

H Glossary of Observation Modes

Table 21. Every observation mode used in the paper. KF = inter-step keyframes (with selection strategy). Audio = transcripts of system audio. Narr. = CU-model visual narration persisted as text. Elem. = DOM element list in the prompt. Traj. = structured prior-step trajectory wrapping. **Standard** and **AOI full** run on all models. The rest are Claude Sonnet 4.6 ablations except where a section indicates a Gemini 3 or open-source diagnostic.

Mode	KF	Audio	Narr.	Elem.	Traj.	Where
Standard	—	—	—	—	✓	Table 2
Minimal prompt	—	—	—	—	—	App. B
+element list	—	—	—	✓	—	App. B
+format (scaffold)	—	—	—	✓	✓	§6
Standard+audio	—	✓	—	—	✓	App. E
Selection variants	variant	—	—	✓	✓	§6.2
+keyframes (visual)	CLIP	—	—	✓	✓	§6.2
+ASR	CLIP	✓	—	✓	✓	§6.2
Scaffold+audio (no KF)	—	✓	✓	✓	✓	§6.4
AOI full	pixel gate	✓	✓	✓	✓	Table 2
Narration discarded	pixel gate	✓	not kept	✓	✓	§6.3
Keyframe probes	modified	✓	✓	✓	✓	App. E

Table 22. DynaCU-Real-Local: Claude Sonnet 4.6 Standard vs. AOI full on 12 real-recording tasks. Both modes tie at 11/12 on this set. The failure is the same task (`R_cast2_pip`) for both.

Mode	Podcast	Meeting	Screencast	Voice	Total / 12
Standard	3/3	3/3	2/3	3/3	**11** (92%)
AOI full	3/3	3/3	2/3	3/3	**11** (92%)

I Implementation Details

Software. Python 3.11, Playwright 1.49 for browser automation, PulseAudio 16.1 for audio I/O, edge-TTS for high-quality speech synthesis, OpenAI CLIP ViT-B/16 for keyframe extraction, Whisper large-v3 for speech transcription, vLLM 0.19.0 for local model serving.

Hardware. All experiments run on a single machine with an NVIDIA RTX PRO 6000 Blackwell GPU (96 GB VRAM), 192 GB RAM. Cloud models (Claude, GPT-5.4, Gemini 2.5 Flash) are accessed via HTTP APIs. Local models (EvoCUA-32B, Fara-7B) are served via vLLM with 4-bit quantization (`bitsandbytes`). Whisper runs on CPU to avoid GPU contention. Running the two local models in the `standard_structured` mode (Section 6.1) required patching vLLM 0.19.0 around an NVML/userspace driver mismatch on our host: the CUDA platform plugin trusts `pynvml.nvmlInit()` alone, and an overly strict free-memory assertion fired when the Whisper service released GPU memory during startup profiling. Both patches are released with the code.

Hyperparameters. Screen capture rate: $\sim$3 Hz. Pixel change threshold (α): 1%. CLIP distance threshold (θ): 0.04. Maximum keyframes per step: 5. Audio capture: a 70 s ring buffer at 16 kHz mono holds rolling history. When the volume gate fires, Whisper is invoked once per step on the new inter-step audio (typically $\sim$3.5 s) extended by a $\sim$3.5 s overlap into the previous interval ($\sim$7 s total) for boundary continuity. Post-action wait: 2.0 s. Maximum steps per task: 15.

J DynaCU-Real-Local Details

The 12 tasks span four sub-domains: 3 podcast (Aesop animal-identification), 3 meeting (Python / Postgres / Rust technical detail), 3 screencast (`git clone` / `pip install` / `npm test` outcome), and 3 voice (yes/no / directions / appointment day). Audio for podcast/meeting/voice tasks is rendered with `espeak` (a formant synthesizer with a deliberately non-neural acoustic profile). Screencast tasks use real `asciinema` v2 cast files generated locally. Source texts are public-domain materials: Aesop's Fables for podcasts, Wikipedia for Python language facts, project documentation for PostgreSQL/MVCC and Rust/tokio. All HTML harnesses, asset-build scripts, and success checks are released with the benchmark.

K Streaming-Baseline Adapter Sanity Check Details

To rule out the explanation that the streaming baselines' weak audio-subset numbers (Section 8) are an artifact of our adapter rather than a model limitation, we evaluate both adapters on a small purely-visual sanity-check set of five tasks across four categories: C (video / static recording), E (live dashboard), F (transient UI, with two tasks: one banner-accept and one cookie-accept subtype), and I (collaborative document). All categories are chosen such that the

Table 23. Per-task adapter sanity-check outcomes for each streaming baseline. Every adapter drives the browser and emits valid tool calls (`click`, `fill`, `type_text`) on every task. Every attempted action lands on the DOM, so the audio-subset failures are content-driven rather than infrastructural. The older adapters score 0/5 task-correct, failing by wrong target (e.g. F-E2 cookies dismissed instead of accepted), hallucinated content (e.g. C-E1), or premature commit (e.g. I-E1), whereas the current GA `gpt-realtime-2` reaches 3/5, closer to but still below the AOI reference, consistent with the audio subset (Table 4).

Baseline	C-E1	E-E1	F-E1	F-E2	I-E1	Total
OpenAI Realtime adapter (gpt-4o backbone)	✗	✗	✗	✗	✗	0
Gemini Live (2.5 flash native-audio)	✗	✗	✗	✗	✗	0
OpenAI `gpt-realtime-2` (GA, native audio)	✓	✓	✓	✗	✗	3
AOI full (Claude Sonnet 4.6) reference	✓	✗	✓	✓	✓	4

answer can be reached without any audio comprehension. The exact set is C-E1, E-E1, F-E1, F-E2, and I-E1. For each baseline, we open one streaming session per task, send the post-action screenshot as a multimodal turn, and translate any returned tool calls into browser actions through the same path used in the main audio-subset evaluation. The success criterion is the same DOM check as in the main benchmark. Table 23 reports the per-task outcome and the modal failure mode where applicable.